



# Trends in College Tuition and student loan debt

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ANLC 752

# Topic

- College tuition has increased rapidly over the last few decades, which has resulted in a growth in student loan debt.
- Evaluate the year, tuition, state, and income level



# Questions



How do tuition increases compare between public, private nonprofit, and private for-profit institutions?



Which states have the highest tuition growth?



Which student groups face the highest debt-to-income ratios?

# DATA SOURCES

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- College Scorecard – Tuition, earnings, debt, institution type

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- IPEDS (NCES) – Historical tuition and enrollment data

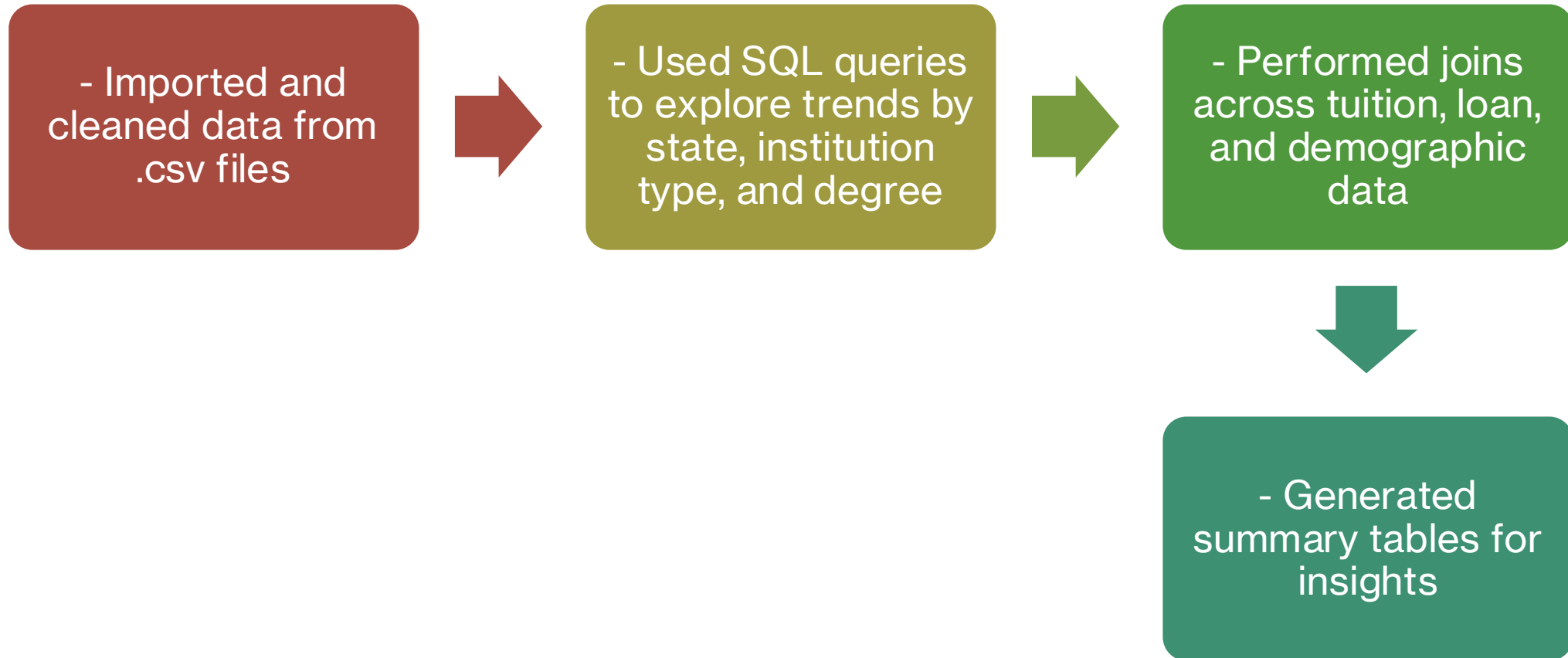
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- Federal Student Aid (FSA) – Loan debt and default rates

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- ACS (U.S. Census) – Demographics by income, education, region

# Loading to Database



# SQL Functions



## Disbursements by State

```
CREATE TABLE disbursements_by_state (  
Location TEXT,  
Grant_Disbursements NUMERIC,  
Grant_Recipients NUMERIC,  
Loan_Disbursements NUMERIC,  
Loan_Recipients NUMERIC,  
Total_Disbursements NUMERIC,  
Total_Recipients NUMERIC,  
Avg_Grant_per_Student NUMERIC(12, 5),  
Avg_Loan_per_Student NUMERIC(12, 5)  
);
```

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Conditional FormattingFormat as TableCell Styles

DeleteFormat

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Add-insAdd-ins

Analyze DataAssistance

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|    | A              | B            | C           | D            | E            | F             | G             | H            | I                    | J | K | L | M | N  | O | P | Q | R | S | T | U |
|----|----------------|--------------|-------------|--------------|--------------|---------------|---------------|--------------|----------------------|---|---|---|---|----|---|---|---|---|---|---|---|
| 1  | location       | grant_disbur | grant_recip | loan_disburs | loan_recipen | total_disburs | total_recipie | avg_grant_pe | avg_loan_per_student |   |   |   |   |    |   |   |   |   |   |   |   |
| 2  | Alabama        | 525.4        | 100.4       | 1245.9       | 102.4        | 1771.3        | 151.8         | 5233.06773   | 12166.99219          |   |   |   |   |    |   |   |   |   |   |   |   |
| 3  | Alaska         | 34.5         | 7.7         | 102          | 8.5          | 136.5         | 13.1          | 4480.51948   | 12000                |   |   |   |   |    |   |   |   |   |   |   |   |
| 4  | Arizona        | 642.2        | 130.6       | 1522.6       | 116.8        | 2164.8        | 193.7         | 4917.30475   | 13035.9589           |   |   |   |   |    |   |   |   |   |   |   |   |
| 5  | Arkansas       | 320.9        | 62.7        | 673          | 60.4         | 994           | 91            | 5118.02233   | 11142.38411          |   |   |   |   |    |   |   |   |   |   |   |   |
| 6  | California     | 4582.4       | 924.8       | 9309.8       | 577          | 13892.1       | 1501.8        | 4955.0173    | 16134.83536          |   |   |   |   |    |   |   |   |   |   |   |   |
| 7  | Colorado       | 400.1        | 82.9        | 1387.5       | 105.5        | 1787.6        | 150.8         | 4826.29674   | 13151.65877          |   |   |   |   |    |   |   |   |   |   |   |   |
| 8  | Connecticut    | 294          | 62.3        | 1216.2       | 89.6         | 1510.2        | 119.3         | 4719.10112   | 13573.66071          |   |   |   |   |    |   |   |   |   |   |   |   |
| 9  | Delaware       | 80.7         | 17.3        | 269.5        | 21.5         | 350.2         | 30            | 4664.73988   | 12534.88372          |   |   |   |   |    |   |   |   |   |   |   |   |
| 10 | District of Co | 45.7         | 8.8         | 246.9        | 12.4         | 292.6         | 16            | 5193.18182   | 19911.29032          |   |   |   |   |    |   |   |   |   |   |   |   |
| 11 | Florida        | 2301.1       | 453.7       | 5271.6       | 375.8        | 7572.7        | 648.5         | 5071.85365   | 14027.67429          |   |   |   |   |    |   |   |   |   |   |   |   |
| 12 | Georgia        | 1385.2       | 271.7       | 3242.6       | 263.3        | 4627.7        | 403.5         | 5098.27015   | 12315.22978          |   |   |   |   |    |   |   |   |   |   |   |   |
| 13 | Hawaii         | 88.8         | 18.6        | 270.7        | 17.9         | 359.5         | 29.9          | 4774.19355   | 15122.90503          |   |   |   |   |    |   |   |   |   |   |   |   |
| 14 | Idaho          | 155.7        | 31.7        | 332.4        | 30.8         | 488.1         | 48.4          | 4911.67192   | 10792.20779          |   |   |   |   |    |   |   |   |   |   |   |   |
| 15 | Illinois       | 1113.5       | 225.5       | 3491.6       | 246.7        | 4605.1        | 377.1         | 4937.91574   | 14153.22254          |   |   |   |   |    |   |   |   |   |   |   |   |
| 16 | Indiana        | 526.2        | 109.2       | 1386.1       | 122.2        | 1912.4        | 179.7         | 4818.68132   | 11342.88052          |   |   |   |   |    |   |   |   |   |   |   |   |
| 17 | Iowa           | 212.5        | 45.1        | 709.6        | 65           | 922.2         | 84.4          | 4711.75166   | 10916.92308          |   |   |   |   |    |   |   |   |   |   |   |   |
| 18 | Kansas         | 229          | 47.2        | 645.1        | 55.6         | 874           | 79.6          | 4851.69492   | 11602.51799          |   |   |   |   |    |   |   |   |   |   |   |   |
| 19 | Kentucky       | 460.9        | 93.9        | 1016.3       | 90.2         | 1477.2        | 140.1         | 4908.41321   | 11267.18404          |   |   |   |   |    |   |   |   |   |   |   |   |
| 20 | Louisiana      | 606          | 116.6       | 1341.8       | 119.9        | 1947.8        | 165.4         | 5197.25558   | 11190.99249          |   |   |   |   |    |   |   |   |   |   |   |   |
| 21 | Maine          | 90.1         | 19.3        | 287.8        | 24.5         | 377.9         | 34.6          | 4668.39378   | 11746.93878          |   |   |   |   |    |   |   |   |   |   |   |   |
| 22 | Maryland       | 492.3        | 101.7       | 1842.1       | 125.4        | 2334.4        | 181.2         | 4840.70796   | 14689.79266          |   |   |   |   |    |   |   |   |   |   |   |   |
| 23 | Massachuset    | 442.1        | 93.9        | 2085.6       | 159.7        | 2527.8        | 205.6         | 4708.20021   | 13059.48654          |   |   |   |   |    |   |   |   |   |   |   |   |
| 24 | Michigan       | 802.4        | 167.7       | 2538.6       | 195.5        | 3341          | 279.8         | 4784.73464   | 12985.16624          |   |   |   |   |    |   |   |   |   |   |   |   |
| 25 | Minnesota      | 386.8        | 83.3        | 1407.9       | 124.3        | 1794.7        | 167.8         | 4643.45738   | 11326.62912          |   |   |   |   | </ |   |   |   |   |   |   |   |

# Aid\_Distribution

```
CREATE TABLE aid_distribution (  
student_group TEXT,  
grant_share NUMERIC,  
work_study_share NUMERIC,  
loan_share NUMERIC,  
family_contribution_share NUMERIC,  
other_share NUMERIC,  
total_share NUMERIC,  
tuition_and_fees NUMERIC,  
grants NUMERIC,  
net_price NUMERIC,  
living_expenses NUMERIC,  
other_expenses NUMERIC,  
total_cost NUMERIC  
);
```

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## Average\_Tuition

```
CREATE TABLE average_tuition (  
    academic_year TEXT,  
    private_nonprofit_four_year NUMERIC,  
    public_four_year NUMERIC,  
    public_two_year NUMERIC  
);
```

average\_tuition

Home Insert Draw Page Layout Formulas Data Review View Automate Developer

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Add-ins

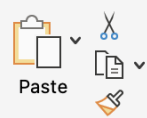
Assistance

Possible Data Loss Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Save As...

|    |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|----|---------------|---------------|---------------|-----------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A1 |               | academic_year |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|    | A             | B             | C             | D               | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| 1  | academic_year | private_nonp  | public_four_y | public_two_year |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2  | 1994-95       | 24840         | 5740          | 2780            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3  | 2004-05       | 33340         | 8530          | 3460            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4  | 2014-15       | 41510         | 12140         | 4430            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 5  | 2024-25       | 43350         | 11610         | 4050            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 6  | 2014-15       | 41510         | 12140         | 4430            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 7  | 2015-16       | 42860         | 12500         | 4510            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 8  | 2016-17       | 43840         | 12660         | 4530            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 9  | 2017-18       | 44480         | 12790         | 4540            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 10 | 2018-19       | 44630         | 12770         | 4540            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 11 | 2019-20       | 45310         | 12830         | 4550            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 12 | 2020-21       | 45240         | 12830         | 4550            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 13 | 2021-22       | 44130         | 12460         | 4410            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 14 | 2022-23       | 42870         | 11800         | 4180            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 15 | 2023-24       | 43030         | 11660         | 4070            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 16 | 2024-25       | 43350         | 11610         | 4050            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 17 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 18 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 19 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 20 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 21 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 22 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 23 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 24 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 25 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 26 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 27 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 28 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 29 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 30 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 31 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 32 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 33 |               |               |               |                 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

# College\_tuition by state

- CREATE TABLE college\_tuition\_by\_state (
  - university TEXT,
  - state TEXT,
  - in\_state\_tuition\_2024\_25 NUMERIC,
  - out\_state\_tuition\_2024\_25 NUMERIC
- );



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## Conditional Formatting



Format  
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## Cell Styles

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Add-ins

Add-ins



## Analyze Data

## Assistance

ⓘ Possible Data Loss Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format.

Save As...

[illegible]

# Cost\_distribution

- CREATE TABLE cost\_distribution (
  - sector TEXT,
  - academic\_year TEXT,
  - tuition NUMERIC,
  - housing\_and\_food NUMERIC,
  - books\_and\_supplies NUMERIC,
  - total\_cost NUMERIC,
  - pct\_tuition NUMERIC,
  - pct\_housing\_and\_food NUMERIC,
  - pct\_other NUMERIC
- );

ⓘ Possible Data Loss Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Save As...

A1    sector

[illegible]

# Data Governance

1

Make sure the datasets you use are available to the public.

2

Make any sensitive personal information anonymous.

3

Verify the sources of the data and accept its limits.

4

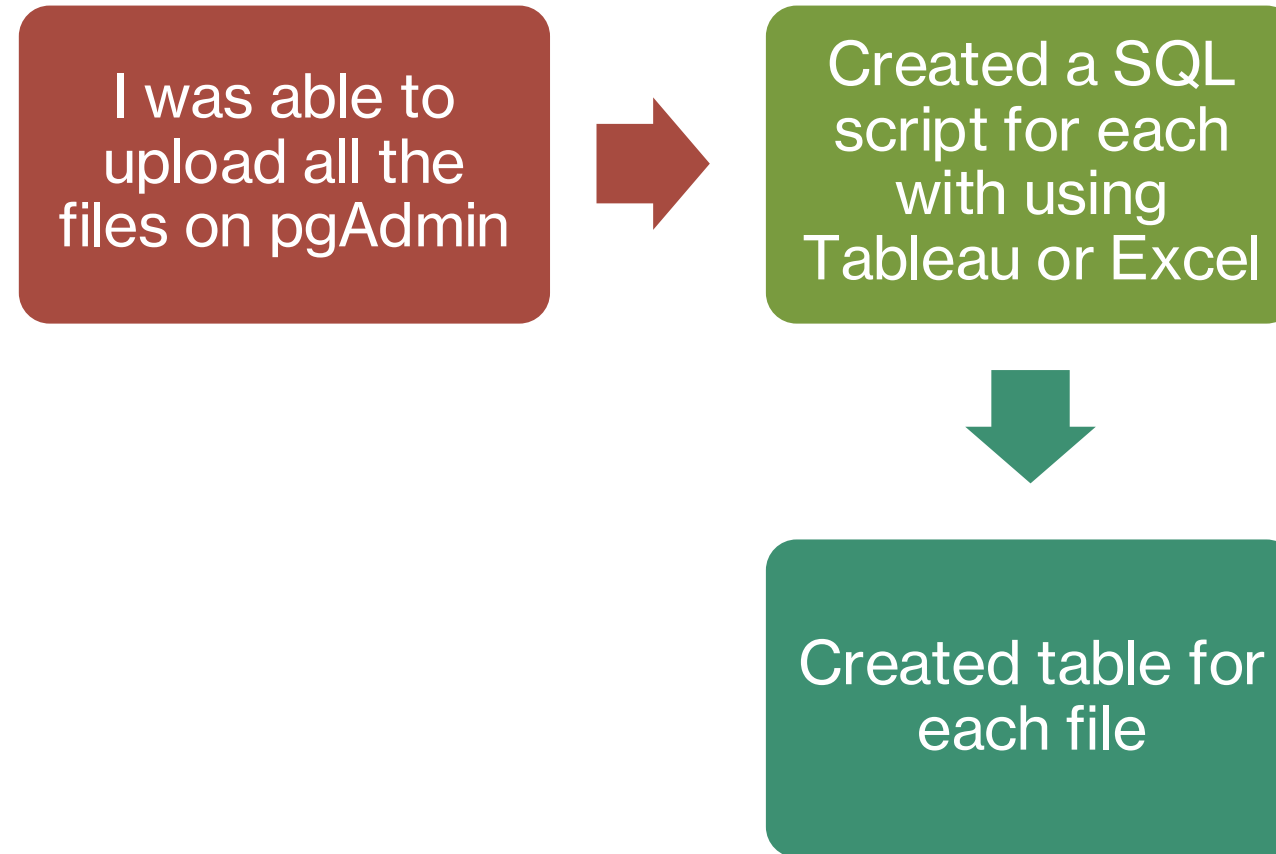
Reduce bias in demographic comparisons

# SQL Script Challenges

Unable to  
upload data  
from Excel via  
SQL script

I'm going to try  
Excel or  
Tableau

# Results



# The End

